Worksheet 2

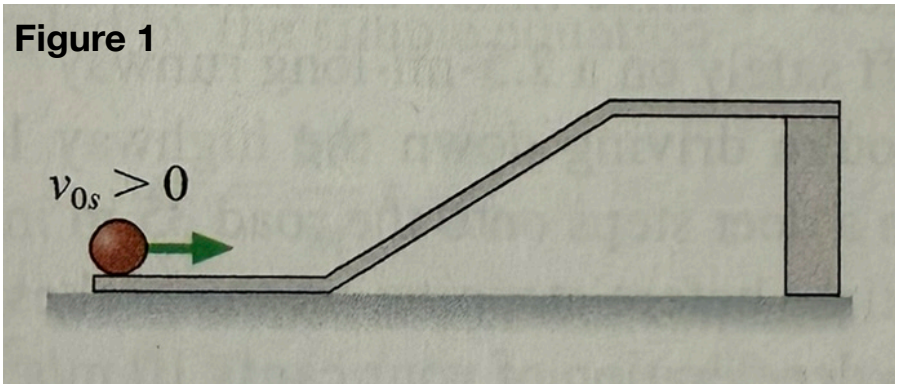
Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).
- Submit before 11:59 PM on Wednesday, September 9, 2026

Problems

This worksheet covers motion in one-dimension, average velocity, and average acceleration.

1. Alan leaves Los Angeles at 8:00 AM to drive to San Francisco, $400\,\text{mi}$ away. He travels at a steady $50\,\text{mph}$. Beth leaves Los Angeles at 9:00 AM and drives a steady $60\,\text{mph}$
   a. Who gets to San Francisco first?
   b. How long does the first to arrive have to wait for the second?



2. A ball rolls along the smooth track shown in figure 1. Each segment of the track is straight, and the ball passes smoothly from one segment to the next without changing speed or leaving the track. Draw three vertically stacked graphs showing position, velocity, and acceleration versus time. Each graph should have the same time axis and the proportions of the graph should be qualitatively correct. Assume that the ball has enough speed to reach the top.
3. Calculate the average velocity of an object that moves from $(x, y) = (0,0)$ to $(3,4)$ in 5 seconds.

   Hint: $\Delta r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

4. A particle moves along the the x-axis according to the relation $x = 6 - 6t + t^2$, where x is measured in meters and t is measured in seconds.
   a. Find the values of x for $t = 1, 2, 3, 4, 5$
   b. Find the values of velocity v for $t = 1, 2, 3, 4, 5$
   c. For each value of t indicate whether the particle is moving toward an increasing or decreasing x.
   d. Is there ever an instant when the velocity is zero?
   e. Is there a time after $t = 5$ when the particle is moving toward decreasing x?